



TOSHKENT SHAHRIDAGI INHA UNIVERSITETI
INHA UNIVERSITY IN TASHKENT

Computer Security

Assignment #1 (Computer Security - Modular Arithmetic)

Name:- _____

ID: _____

Submission Date: 3 /03/ 2026 [Tuesday] (23:30 Hr)

NOTE :

1. Take the **print of this sheet** and solve on it. Computer typed solution are not acceptable (**only handwritten** allowed)
2. Submissions not conforming to the given format will not be evaluated.

Using Fermat's Little Theorem Solve Following questions **Number 1-3:**

1. $4^{225} \bmod 13$.

Solution:

Name:- _____

ID: _____

2. Solve the congruence
 $X^{73} \equiv 4 \pmod{37}$. Find x .

Solution:

3. $7^{1013} \pmod{93}$

Solution:

4. Using Extended Euclidean Algorithm (EEA) find $135^{-1} \pmod{61}$

Solution:

Q	A	b	r	x_1	x_2	X

Name:- _____

ID: _____

5. Calculate $\Phi(n)$ for $n=1200$ and $n=2008$.

Solution: $\Phi(1200)$	Solution: $\Phi(2008)$

6. Decrypt the following ciphertext, which is made with the Playfair cipher

key : CIPHER

Ciphertext: FAFIDLKGVQAGIDAFTUDWUROEPUBY

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Name:- _____

ID: _____

7. Decrypt the ciphertext “**ENOEXDOIPWWAXUISTELPXCTMSRUOXAOFN**” using a rail fence cipher with key 5.

8. Ciphertext is F and key is (3, 1) find plaintext in AFFINE CIPHER.

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9. Decrypt ciphertext ZB using HILL CIPHER key matrix is

$$\begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$$